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MULTIMODAL INTERACTIVE TECHNOLOGIES AND APPLICATIONS

Project Concept

Botanical Garden

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1 Vision

MITA (Multimodal Interactive Tour Assistant) is a browser-based, immersive botanical garden combining **WebVR panoramic exploration** with a **conversational, voice-driven assistant**. Users walk through 360° panoramic scenes, point at plants in 3D and *talk* to the system in natural language, turning a traditional static-label exhibit into a fluid multimodal dialogue between user, environment and content.

2 Characterizing Features

- **Immersive 3D environment.** Built on **A-Frame (WebVR)** and rendered in-browser. Five panoramic scenes (`scene-1...scene-5`) are linked by clickable arrow portals; plant specimens (*Lavandula*, *Nephrolepis*, *Santolina*, *Opuntia*...) are positioned `glTF` models inspectable from any angle.
- **Multimodal input.** Three integrated channels: (i) *gaze / pointer raycasting* to select a plant in 3D; (ii) *speech (STT)* via the Web Speech API for hands-free queries (“*What is this used for?*”, “*Compare it with the lavender.*”); (iii) *click / touch fallback* for accessibility and noisy environments.
- **Multimodal output.** Coordinated combination of a *spatial highlight* (glow / outline) on the selected plant, *adaptive 2D HUD cards* (botanical, medicinal, comparison, fallback), and *TTS narration* so the user never needs to leave the VR view to read.
- **LLM-driven dialogue manager.** A minimal Node.js HTTP server exposes JSON endpoints (`/api/parse-intent`, `/api/generate-info`, `/api/compare...`) that perform LLM-backed **intent parsing** (`query_info`, `query_attribute`, `compare`) and **slot extraction** (`medicinal`, `botanical`, `toxicity`, `drought`, `feng_shui`), interpreted against the selected plant, dialogue history and the structured `plants.json` database. The backend is intentionally stateless; each request carries all context needed for that turn so the interaction loop remains available even when the LLM service is offline.
- **Client-side interaction state.** A finite set of response categories (`response`, `comparison`, `fallback`, `disambiguate`, `clarify`, `clarify_retry`, `error`) and **client-side state** centralised in `src/app/state.js` — tracking the selected plant, visited plant ids, pending ambiguity candidates and dialogue history — allow the assistant to reason contextually across turns without any persistent server session.
- **Robust disambiguation.** When raycasting returns overlapping plants, candidates are ranked by depth, labelled *A / B / C* and announced verbally; the user resolves by voice (“*the one in front*”, “*A*”) or by clicking the overlay, with up to two clarification retries before falling back to pointer selection.

3 Target Users and Why It Is Multimodal

The system targets **casual museum visitors, botany students and accessibility-oriented users** exploring a botanical collection on desktop, tablet or VR headset. Rather than reading long labels, users ask focused questions (“*Can I keep this around my cat?*”, “*How does it compare*

to the cactus?”) and receive concise spoken answers paired with a visual card. MITA is multimodal **by design**: every interaction loop fuses at least two input modalities (point + speak) and three output modalities (3D highlight + card + voice), and degrades gracefully — if STT fails the user can click; if the LLM is uncertain a structured A/B overlay takes over; if a slot is missing, a *fallback card* proposes alternatives. This redundancy keeps the experience fluent across users, devices and ambient conditions.